

Bounds on System Reliability by Linear Programming

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Abstract: Bounds on system probability in terms of marginal or joint component probabilities are of interest when exact solutions cannot be obtained. Currently, bounding formulas employing unicomponent probabilities are available for series and parallel systems, and formulas employing bi- and higher-order component probabilities are available for series systems. No theoretical formulas exist for general systems. It is shown in this paper that linear programming (LP) can be used to compute bounds for any system for any level of information available on the component probabilities. Unlike the theoretical bicomponent and higher-order bounds, the LP bounds are independent of the ordering of the components and are guaranteed to produce the narrowest possible bounds for the given information. Furthermore, the LP bounds can incorporate any type of information, including an incomplete set of component probabilities or inequality constraints on component probabilities. Numerical examples involving series, parallel and general structural systems are used to demonstrate the methodology.

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Introduction

A system, in general, is a set of possibly interconnected or interdependent components, such that the state of the system depends on the states of its constituent components. For many systems, the system state can be expressed as a Boolean or logical function (consisting of unions and intersections) of the component states. The system reliability, i.e., the probability that the system is in a particular functioning state, or its complement, the system failure probability, can then be expressed as the probability of the Boolean function of the component states that produce that system state. Computation of this probability, however, is extremely difficult for many systems, particularly when there is dependence between the component states. Because of this difficulty, attempts have been made to derive bounds on the system failure probability by use of individual component probabilities or joint probabilities of small sets of the components. Available formulas for such bounds are primarily restricted to series and parallel systems (see the next section).

In this paper, we propose a method for computing bounds on the system failure probability by use of linear programming (LP). The idea of using LP to compute bounds on system reliability was first explored by Hailperin (1965). Kounias and Marin (1976) used the approach to examine the accuracy of some theoretical bounds. Since the number of variables in the LP problem increases exponentially with the number of system components,

these early attempts were abandoned as they were computationally too demanding for the computers existing at that time. Later, specialized versions of this approach were employed in fields such as operations research (Prékopa 1988). However, it appears that this approach has never been used in the field of structural reliability, which is the application focus of the present study. With the enormous increase in the speed and capacity of computers in recent years, we believe the LP approach is viable and worthy of a reconsideration for all system reliability problems.

LP has previously been used in the context of structural system reliability by Nafday et al. (1987) and Corotis and Nafday (1989). However, they used LP for the purpose of identifying the most critical failure mode for a structural system, not for computing the system probability. The present paper shows that LP is useful in structural system reliability analysis in more than one way.

The LP bounds developed in this paper are applicable for any type of system and any level of information regarding the component probabilities. Equally important, these bounds are the narrowest possible bounds that one can obtain for a system for any specified information regarding the component probabilities.

The proposed method for computing bounds on the system reliability is demonstrated by application to three example structural systems. The first example, a statically determinate truss, is a series system. Through this example, it is shown that existing formulas for reliability bounds of series systems employing bi- or tri-component probabilities may not yield the narrowest possible bounds for the given information. The second example represents a “Daniels” system consisting of a bundle of ideally brittle wires with random capacities. This is a parallel system for which an exact reliability solution is available. For this example, comparisons are made between the exact solution, existing bounds based on uni- and bi-component probabilities, and the LP bounds based on bicomponent probabilities. The last example is for a structural system having both ductile and brittle components. This is a general system (neither series nor parallel) and is formulated by the use of cut sets. No formulas are currently available for computing bounds for such systems. The LP bounds for this system are com-

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puted by the use of uni- and bi-component probabilities and are compared with the “exact” Monte Carlo solution.

Formulation of System Reliability

Consider a system having n components. Let E_{system} denote a particular system state of interest (e.g., the state of failure relative to a prescribed performance criterion) and $\mathbf{E}_i = (E_{i1}, E_{i2}, \dots)$, $i = 1, \dots, n$, denote vectors of the component states such that E_{im} = the event that component i is in its m th state. In general one can write

$$E_{\text{system}} = L(\mathbf{E}_1, \dots, \mathbf{E}_n) \quad (1)$$

where $L(\cdot)$ denotes a logical function involving unions and intersections of the component events or their complements. Specific cases of this function are described below. For the sake of simplicity of the notation, and without loss of generality, in the following we will consider two-state components, where $\mathbf{E}_i$ can be written as $\mathbf{E}_i = (E_i, \bar{E}_i)$, where the superposed bar indicates the complement of an event. In that case, Eq. (1) simplifies to

$$E_{\text{system}} = L(E_1, \bar{E}_1, \dots, E_n, \bar{E}_n) \quad (2)$$

For easy visualization, it is useful to think of E_i ($\bar{E}_i$) as the state of failure (survival) of component i and E_{system} as the state of failure of the system.

Mathematically speaking, a series system is one in which $L(\cdot)$ includes only union operations (over all or a subset of the component events), i.e.

$$E_{\text{series system}} = \bigcup_i E_i \quad (3)$$

In this case, the system fails if any of its components fail. A parallel system is characterized by intersection operations, i.e.

$$E_{\text{parallel system}} = \bigcap_i E_i \quad (4)$$

In this case, the system fails if all its components fail. More generally, the system function $L(\cdot)$ may include both union and intersection operations. In that case, two alternative formulations are possible:

$$E_{\text{system}} = \bigcup_k \bigcap_{i \in C_k} E_i \quad (5)$$

$$E_{\text{system}} = \bigcap_l \bigcup_{i \in L_l} E_i \quad (6)$$

The formulation in Eq. (5) is in terms of “cut sets,” i.e., sets of component states E_i , $i \in C_k$, whose joint realizations constitute realizations of the system state E_{system} . In this expression, C_k denotes the set of component state indices that constitute the k th cut set. The system in this case is represented by a series of parallel subsystems. The formulation in Eq. (6) is in terms of “link sets,” i.e., sets of complementary component states $\bar{E}_i$, $i \in L_l$, whose joint realizations constitute realizations of the complementary system state $\bar{E}_{\text{system}}$. In this expression L_l denotes the set of component state indices that constitute the l th link set. The system in this case is represented by a parallel of series subsystems. The form in Eq. (6) is obtained by use of De Morgan’s rule.

Computing the probability for any of the system events given above is a daunting task when the component events are statistically dependent. One in general needs to know the probabilities of intersections of all combinations of component states. For

example, for the series system, using the inclusion-exclusion rule, one can write

$$P(E_{\text{series system}}) = P\left(\bigcup_i E_i\right) = \sum_i P(E_i) - \sum_{i < j} P(E_i E_j) + \sum_{i < j < k} P(E_i E_j E_k) - \dots \quad (7)$$

Similar expressions can be developed for the general system formulations in Eqs. (5) and (6). When the component events are statistically dependent, computation of the probabilities of event intersections is difficult for a large number of components. Because of this, there has been continued interest in developing bounds on the system probability that employ the marginal component probabilities, $P_i = P(E_i)$, and joint probabilities of small sets of component events, i.e., bicomponent probabilities, $P_{ij} = P(E_i E_j)$, $i < j$, tricomponent probabilities, $P_{ijk} = P(E_i E_j E_k)$, $i < j < k$, etc.

For series structural systems, probability bounds using uni-component probabilities were derived by Boole (1854):

$$\max_i P_i \leq P\left(\bigcup_{i=1}^n E_i\right) \leq \min\left(1, \sum_{i=1}^n P_i\right) \quad (8)$$

These are the narrowest possible bounds when only the unicomponent probabilities are known (Fréchet 1935). However, these bounds are often too wide to be of practical use. Efforts made by Kounias (1968); Hunter (1976); and Ditlevsen (1979) led to the following widely used bounds for series systems that involve the uni- and bi-component probabilities:

$$P_1 + \sum_{i=2}^n \max\left(0, P_i - \sum_{j=1}^{i-1} P_{ij}\right) \leq P\left(\bigcup_{i=1}^n E_i\right) \leq P_1 + \sum_{i=2}^n (P_i - \max_{j < i} P_{ij}) \quad (9)$$

These bounds depend on the ordering of the component events, and the order that maximizes the lower bound and the order that minimizes the upper bound are not necessarily the same. In order to obtain the narrowest bounds, one must consider all the possible $n!$ ordering alternatives, since there does not exist an ordering rule that guarantees this result. Furthermore, there is no proof that the narrowest bounds obtained from this formula are the narrowest possible bounds for the specified information. Through a counterexample, we will later show that this, in fact, is not the case. In the remainder of this paper, we will refer to the above bounds as “KHD bounds.”

Zhang (1993) generalized the concept of the KHD bounds for series systems by including joint probabilities of larger sets of component events, i.e., tricomponent probabilities, P_{ijk} , quadri-component probabilities, P_{ijkl} , etc. The bounding formulas for up to tricomponent probabilities are

$$P\left(\bigcup_{i=1}^n E_i\right) \geq P_1 + P_2 - P_{12} + \sum_{i=3}^n \max\left(0, P_i - \sum_{j=1}^{i-1} P_{ij} + \max_{k \in (1, 2, \dots, i-1)} \sum_{\substack{j=1 \\ j \neq k}}^{i-1} P_{ijk}\right) \quad (10a)$$

$$P\left(\bigcup_{i=1}^n E_i\right) \leq P_1 + P_2 - P_{12} + \sum_{i=3}^n [P_i - \max_{\substack{k \in \{2,3,\dots,i-1\} \\ j < k}} (P_{ik} + P_{ij} - P_{ijk})] \quad (10b)$$

These bounds inherit the order-dependency problem of KHD bounds.

For parallel systems, bounds based on unicomponent probabilities derived by Boole (1854) are

$$\max\left(0, \sum_{i=1}^n P_i - (n-1)\right) \leq P\left(\bigcap_{i=1}^n E_i\right) \leq \min_i P_i \quad (11)$$

Fréchet (1935) has shown that these are the narrowest possible bounds based on the unicomponent probabilities alone. Unfortunately, these bounds are too wide to be of practical use for most systems. Theoretical bounds using bi- or tri-component probabilities do not exist for parallel systems. However, one can use the De Morgan's rule $\overline{\bigcap E_i} = \bigcup \overline{E_i}$ to convert the complement of the parallel system to a series system involving the complementary component events, and then use Eq. (9) or (10). We will use this approach for an example parallel system.

For general systems, no theoretical bounding formulas exist. One can of course represent the system as a series system of parallel subsystems, and then combine Eq. (9) or (10) with Eq. (11) to develop relaxed unicomponent bounds, or represent the system as a parallel system of series subsystems, and then combine Eq. (11) with Eq. (9) or (10) to develop relaxed bi- or tri-component bounds. However, these usually lead to unacceptably wide bounds.

Linear Programming

Linear programming (LP) solves the problem of minimizing (maximizing) a linear function, whose variables are subject to linear equality or inequality constraints. The first appearance of LP goes back to J. B. Fourier's work in 1824, but it became practical after G. B. Dantzig developed the simplex method in 1947 (Dantzig 1951). Since then, together with dramatic improvements in computing technology, many powerful algorithms have been developed and a profound mathematical understanding of the problem has been gained. The compact formulation of LP appropriate for our analysis has the form

$$\text{minimize (maximize)} \quad \mathbf{c}^T \mathbf{p} \quad (12a)$$

$$\text{subject to} \quad \mathbf{a}_1 \mathbf{p} = \mathbf{b}_1 \quad (12b)$$

$$\mathbf{a}_2 \mathbf{p} \geq \mathbf{b}_2 \quad (12c)$$

In the above, $\mathbf{p} = (p_1, p_2, \dots)$ is the vector of "decision" or "design" variables, $\mathbf{c}^T \mathbf{p}$, where $\mathbf{c}$ is a vector of coefficients, is the linear "objective" or "cost" function, and $\mathbf{a}_1$, $\mathbf{b}_1$, $\mathbf{a}_2$, and $\mathbf{b}_2$ are coefficient matrices and vectors that respectively define equality and inequality constraints. In Eq. (12c), the inequality between the vectors must be interpreted component-wise. A vector $\mathbf{p}$ is called feasible if it satisfies all the constraints. The solution of the LP problem is a feasible $\mathbf{p}$ that minimizes (maximizes) the objective function.

The following theorem plays a key role in understanding LP. The detailed proof can be found in Bertsimas and Tsitsiklis (1997).

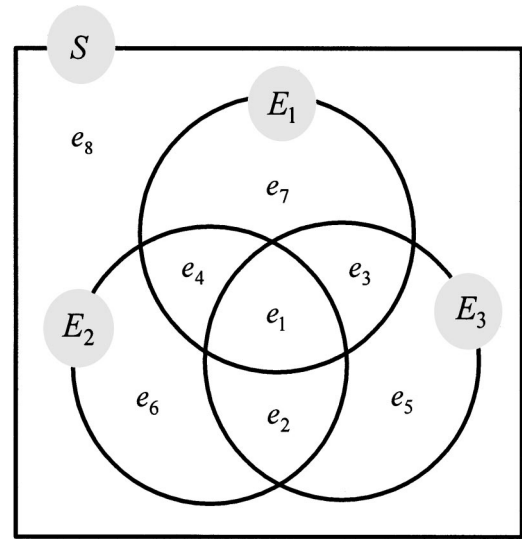


Fig. 1. Basic MECE events e_i for a three-event sample space

Theorem 1. Consider the linear programming problem of minimizing (maximizing) $\mathbf{c}^T \mathbf{p}$ over a polyhedron P defined by the imposed constraints. Suppose that P has at least one extreme point (vertex). Then, either the optimal cost is equal to $-\infty$ (∞), or there exists an extreme point which is optimal.

The theorem implies that the optimal solution of LP, if it exists, is located at one of the extreme points (vertices) of the polyhedron defined by the given linear constraints. This fact led to the development of the simplex algorithm (Dantzig 1951), which moves from one vertex to another under a certain pivoting rule, until the requirements for the optimal solution are met. Since the simplex algorithm appeared, LP has flourished and numerous algorithms (interior point method, ellipsoid method, etc.) have been developed, dramatically increasing our ability to solve large-scale problems. Useful information and on-line LP solvers can be found at the Internet web site <http://www-neos.mcs.anl.gov>.

Bounds by Use of Linear Programming

Hailperin (1965) divided the sample space of the component events into 2^n mutually exclusive and collectively exhaustive (MECE) events, each consisting of a distinct intersection of the component events E_i and their complements $\overline{E_i}$, $i = 1, \dots, n$. We call these the basic MECE events and denote them by e_i , $i = 1, \dots, 2^n$. For example, in the case of $n = 3$ component events, one finds the $2^3 = 8$ basic MECE events to be $\{e_1 = E_1 E_2 E_3, e_2 = \overline{E_1} E_2 E_3, e_3 = E_1 \overline{E_2} E_3, e_4 = E_1 E_2 \overline{E_3}, e_5 = \overline{E_1} \overline{E_2} E_3, e_6 = \overline{E_1} E_2 \overline{E_3}, e_7 = E_1 \overline{E_2} \overline{E_3}, \text{ and } e_8 = \overline{E_1} \overline{E_2} \overline{E_3}\}$ (see Fig. 1). Let $p_i = P(e_i)$, $i = 1, \dots, 2^n$, denote the probabilities of the basic MECE events. These probabilities serve as the design variables in the LP problem to be formulated.

According to the basic axioms of probability, the probabilities $\mathbf{p} = (p_1, \dots, p_{2^n})$ are subject to the following linear constraints:

$$\sum_{i=1}^{2^n} p_i = 1 \quad (13a)$$

$$p_i \geq 0, \quad \forall i \quad (13b)$$

Table 1. Coefficients c_i of the Object Functions $\mathbf{c}^T \mathbf{p}$ for Three-Component Systems

System event	Basic MECE Events							
	$e_1 = E_1 E_2 E_3$	$e_2 = \bar{E}_1 E_2 E_3$	$e_3 = E_1 \bar{E}_2 E_3$	$e_4 = E_1 E_2 \bar{E}_3$	$e_5 = \bar{E}_1 \bar{E}_2 E_3$	$e_6 = \bar{E}_1 E_2 \bar{E}_3$	$e_7 = E_1 \bar{E}_2 \bar{E}_3$	$e_8 = \bar{E}_1 \bar{E}_2 \bar{E}_3$
	Design Variables							
	p_1	p_2	p_3	p_4	p_5	p_6	p_7	p_8
$E_1 \cup E_2 \cup E_3$	1	1	1	1	1	1	1	0
$E_1 E_2 E_3$	1	0	0	0	0	0	0	0
$E_1 E_2 \cup E_3$	1	1	1	1	1	0	0	0
$(E_1 \cup E_2)(\bar{E}_2 \cup E_3)$	1	1	1	0	0	0	1	0

The constraint (13a) is analogous to (12b) with $\mathbf{a}_1$ being a row vector of 1's and $\mathbf{b}_1 = 1$, whereas Eq. (13b) is analogous to Eq. (12c) with $\mathbf{a}_2$ being an identity matrix of size 2^n and $\mathbf{b}_2$ a 2^n -vector of 0's.

Due to mutual exclusivity of the basic MECE events, the probability of any subset made of these events is the sum of the corresponding probabilities. In particular, the probability of any component event E_i is the sum of the probabilities of the basic MECE events that constitute that component event. For example, for the system with three components mentioned above

$$\begin{aligned} P(E_1) &= P_1 = p_1 + p_3 + p_4 + p_7 \\ P(E_2) &= P_2 = p_1 + p_2 + p_4 + p_6 \\ P(E_3) &= P_3 = p_1 + p_2 + p_3 + p_5 \end{aligned} \quad (14)$$

Similarly, the probability of any intersection of the component events is given as the sum of the probabilities of the basic MECE events that constitute the intersection event. For example, the bi-component probabilities of the three component example are given by

$$\begin{aligned} P(E_1 E_2) &= P_{12} = p_1 + p_4 \\ P(E_1 E_3) &= P_{13} = p_1 + p_3 \\ P(E_2 E_3) &= P_{23} = p_1 + p_2 \end{aligned} \quad (15)$$

More generally, we can write

$$P(E_i) = P_i = \sum_{r: e_r \subseteq E_i} p_r \quad (16a)$$

$$P(E_i E_j) = P_{ij} = \sum_{r: e_r \subseteq E_i E_j} p_r \quad (16b)$$

$$P(E_i E_j E_k) = P_{ijk} = \sum_{r: e_r \subseteq E_i E_j E_k} p_r \quad (16c)$$

$$\text{etc.} \quad (16d)$$

In most system reliability problems, the uni-, bi- and sometimes tri-component probabilities are known or can be computed. In that case, the above expressions provide linear equality constraints on the variables $\mathbf{p}$ in the form of Eq. (12b) with $\mathbf{a}_1$ a matrix having elements of 0 or 1 and $\mathbf{b}_1$ a vector listing the known component probabilities. If, instead, inequality constraints on component probabilities are given, such as $P_i \leq 0.1$, $0.01 \leq P_{ij} \leq 0.02$, or $P_i \leq P_j$, then the above expressions provide linear inequality constraints on the variables $\mathbf{p}$ in the form of Eq. (12c).

Any Boolean function of the component events can also be considered as being composed of a subset of the basic MECE events. It follows that the probability of the system event E_{system}

can be written in the form $P(E_{\text{system}}) = \mathbf{c}^T \mathbf{p}$, where $\mathbf{c}$ is a vector whose elements are either 0 or 1. Table 1 lists the elements of the vector $\mathbf{c}$ for example systems with $n=3$ components. Included are series, parallel, and general systems, the latter represented by both cut-set and link-set formulations. It is seen that in all cases the system probability is a linear function of $\mathbf{p}$.

It is clear from the above analysis that the system reliability problem can be cast in the form of an LP problem. The probabilities of the basic MECE events represent the design variables, the system probability defines the linear objective function, and information given in terms of individual or joint component probabilities define the linear equality or inequality constraints. Additional linear constraints are imposed by the axioms of probability. The lower bound of the system probability is obtained as the minimum of the objective function, and the upper bound is obtained as the maximum of the objective function. For a system with n component events, the number of design variables is 2^n ; one equality and 2^n inequality constraints result from the probability axioms (13a) and (13b), respectively [note that Eq. (13b) is automatically satisfied by only searching in the non-negative domain of the design variable space], n equality or inequality constraints result from knowledge of unicomponent probabilities or bounds thereof as in Eq. (16a), $n!/[2!(n-2)!]$ equality or inequality constraints result from knowledge of bicomponent probabilities or bounds thereof as in Eq. (16b), and so on. Obviously the size of the LP problem quickly grows with the number of component events. This issue is further discussed below.

Hailperin (1965) has shown that the LP problem described above gives the best (narrowest) possible bounds on the system probability for any given information on the individual or joint component probabilities, as long as the given constraints are feasible, i.e., such that the polyhedron defined by the constraints is not empty. This is also evident from the following argument: If the polyhedron is not empty, i.e., the given constraints are feasible, the polyhedron will have at least one extreme point. Then, according to Theorem 1 in "Linear programming," the optimal cost is either equal to $-\infty(+\infty)$, or there exists an extreme point which is optimal. In our LP problem, the design variables p_i are bounded between 0 and 1 and vector $\mathbf{c}$ in the objective function contains only 0's and 1's. It follows that for any feasible $\mathbf{p}$ the cost is finite and, therefore, there must exist an optimal solution, i.e., a global minimum (maximum) value, under the given constraints. Thus the bounds obtained by LP are not only true bounds but also the narrowest bounds that one can obtain by use of the given information (constraints).

The bounds by LP have a number of advantages over the existing theoretical bounds. First, LP is guaranteed to provide the narrowest possible bounds if a feasible solution exists for the given constraints. As we will see later, this is not the case for the

theoretical bounds for series systems based on the bi- or tri-component probabilities, even for the best ordering of the component events. (Note that the LP formulation is independent of the ordering of the component events.) Second, the LP formulation is uniformly applicable to all systems, including general systems characterized by unions and intersections of component events. As mentioned earlier, existing formulas for bounds based on bi- and tri-component probabilities are primarily restricted to series systems. Third, the LP formulation can incorporate general forms of information about the component probabilities. Specifically, any linear equality or inequality expression involving uni- or multi-component probabilities can be used. For example, information conveyed in the form of equality expressions such as $P_i = 0.01$, $P_j + P_k = 0.02$, and $P_{kl} = 0.001$, or inequality expressions such as $0.01 \leq P_i \leq 0.02$, $P_i \leq P_j$, or $P_{ij} \leq P_{kl}$ are easily incorporated. Note that with the LP approach it is not necessary to have the complete set of probabilities for all components at a particular, e.g., uni-, bi- or tri-, level. Any partial set of the component probabilities can be used.

The main drawback of the LP approach is that the size of the problem increases exponentially with the number of component events. This drawback may have been the main reason for not pursuing this approach when it was first suggested by Hailperin in 1965. However, with the enormous speed and memory capacity of today's computers, this purely computational problem is not as acute. The size of the LP problem is usually determined by the number of design variables and the number of constraints. For our LP problem, the number of constraints, which depends on the available information on component probabilities, is usually not critical. However, the number of design variables, which is 2^n for a system with n components, exponentially grows with the number of components. The fact that, for our formulation, the design variables are all bounded within the interval (0,1) and all coefficients in the objective function or the constraint equations are either 0 or 1 provides a considerable advantage. Problems with $n \leq 17$ (about 100,000 design variables) can be solved with ordinary LP solvers on a PC. For larger problems, computers with larger memory and/or parallel computing may be necessary. There are a number of advanced LP algorithms for such large problems [see Chap. 6 in Bertsimas and Tsitsiklis (1997)]. The discussion of these algorithms is beyond the scope of this paper. With rapidly advancing speed and capacity of computers, these purely computational issues may not be a major hindrance in the application of the approach to many systems' reliability problems, even with large n . We note, in passing, that for $n = 17$ the number of orderings of the component events is $17! = 3.56 \times 10^{14}$. In using the KHD or the higher order bounds by Zhang for series systems, obviously it will not be possible to check all the possible orderings of the component events. In that case, bounds computed by these theoretical formulas can potentially be far from the narrowest possible bounds.

It is possible to conceive of ways to reduce the size of the LP problem at the expense of relaxing the bounds. One way to do this is to assign fixed values to a subset of the design variables p_i that are within the feasible domain. The resulting LP problem with the remaining "free" set of design parameters will produce a suboptimal solution pair (maximum and minimum). These solutions truly bound the system probability, but they may not be the narrowest possible bounds for the given information. In doing so, it would be best to select those design variables that are expected to have small values and, therefore, make small contributions to the objective function. These variables are easily identified by examining equality constraints such as Eqs. (16a), (16b), and (16c).

Suppose P_i is known to be much smaller than the other component probabilities. According to Eq. (16a), the set of p_r 's constituting P_i must also be small. Therefore these variables are good candidates for assigning fixed values that satisfy Eq. (16a). A similar reduction can be exercised by considering bi- and tri-component probabilities that are small in relation to other bi- and tri-component probabilities.

In summary, while the LP approach may become computationally demanding for systems with a large number of component events (say greater than 17), it has the following important advantages: (1) it provides the narrowest possible bounds for any given level of information on component probabilities, (2) it is independent of the ordering of the component events, (3) it can incorporate general forms of information about component events, (4) it is uniformly applicable to all types of systems, and (5) general-purpose software is widely available for solving the problem. Furthermore, computational limitations of the approach will diminish with increasing speed and memory capacity of computers.

In the following we present examples selected from structural system reliability analysis to demonstrate the above advantages of the LP formulation.

Example Applications

Three numerical examples are considered in this section. These are taken from the field of structural reliability. The first example deals with a statically determinate truss with seven members, which is modeled as a series system with $n = 7$ components. The uni-, bi- and tri-component bounds are estimated by LP and are compared with the corresponding theoretical bounds. It is found that in some cases the theoretical formulas do not provide the narrowest possible bounds, even when all possible component ordering choices are considered. The second example deals with the failure probability of a Daniels' system, a bundle of brittle wires. The system is modeled as a parallel system and the uni- and bi-component probabilities are computed by order statistics. Unicomponent bounds on the system probability are computed by Boole's formula and bicomponent bounds are computed by LP and by the KHD formula. The results are compared with the exact solution given by Daniels. The final example deals with a structural system having both ductile and brittle members. The system is modeled through a cut-set formulation. Uni- and bi-component bounds are computed by LP and compared with the "exact" solution obtained by Monte Carlo simulation. The simplex algorithm implemented in Matlab Optimization Toolbox is used to solve all the LP problems.

Example 1: Truss as a Series System

Consider the truss in Fig. 2, which is adopted from Ditlevsen (1979). Since this is a statically determinate structure, the failure of any member constitutes failure of the truss. Therefore the truss is a series system with its members representing the components. Let L denote the load acting on the truss. Neglecting the buckling failure mode, let X_i , $i = 1, 2, \dots, 7$, denote the tensile/compressive strength of the i th member. Based on the distribution of internal forces shown in Fig. 2, the failure events of the individual members are $E_i = \{X_i \leq L/(2\sqrt{3})\}$ for $i = 1$ and 2, and $E_i = \{X_i \leq L/\sqrt{3}\}$ for $i = 3, 4, \dots, 7$. Suppose the load has the deterministic value $L = 100$ and the member strengths X_i , $i = 1, 2, \dots, 7$, are jointly normally distributed random variables with X_1 and X_2 having means 100 and standard deviations 20 and $X_3 - X_7$ having

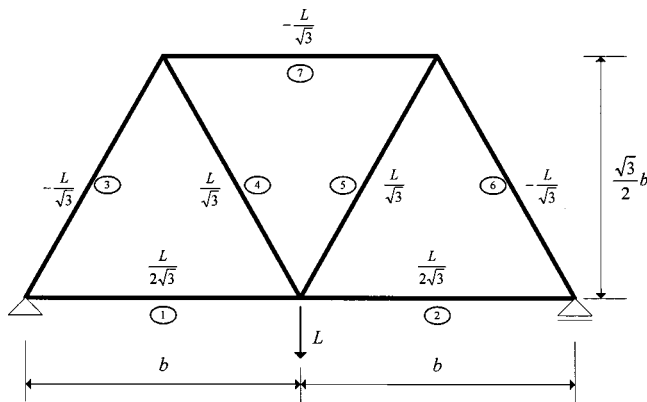


Fig. 2. Statically determinate truss as a series system

means 200 and standard deviations 40. Under these conditions, the members have equal probabilities of failure given by

$$P_i = P(E_i) = \Phi\left(\frac{100/\sqrt{3} - 200}{40}\right) = 1.88 \times 10^{-4} \quad i = 1, 2, \dots, 7 \quad (17)$$

where $\Phi(\cdot)$ denotes the standard normal cumulative probability. Further, suppose X_i 's have a Dunnett–Sobel (DS) class correlation matrix, which is specified as $\rho_{ij} = r_i r_j$ for $i \neq j$ and $\rho_{ii} = 1$. In that case, the m -component joint failure probabilities can be computed by the one-dimensional integral (Dunnett and Sobel 1955)

$$P_{12 \dots m} = P\left(\bigcap_{i=1}^m E_i\right) = \Phi_m(u_1, \dots, u_m; \mathbf{R}) \\ = \int_{-\infty}^{\infty} \left[\varphi(t) \prod_{i=1}^m \Phi\left(\frac{u_i - r_i t}{\sqrt{1 - r_i^2}}\right) \right] dt \quad (18)$$

where $\Phi_m(u_1, \dots, u_m; \mathbf{R})$ is the m -variate standard normal cumulative probability function with correlation matrix $\mathbf{R} = [\rho_{ij}]$ at co-ordinates $u_i = (100/\sqrt{3} - 200)/40$ and $\varphi(\cdot)$ denotes the one-dimensional standard normal probability density function.

First consider the case $r_1 = 0.90$, $r_2 = 0.96$, $r_3 = 0.91$, $r_4 = 0.95$, $r_5 = 0.92$, $r_6 = 0.94$, and $r_7 = 0.93$. The bi- and tri-component probabilities, P_{ij} and P_{ijk} , respectively, computed using Eq. (18) are ($\times 10^{-5}$) $P_{12} = 5.73$, $P_{13} = 4.35$, $P_{14} = 5.42$, $P_{15} = 4.59$, $P_{16} = 5.13$, $P_{17} = 4.85$, $P_{23} = 6.08$, $P_{24} = 7.79$, $P_{25} = 6.47$, $P_{26} = 7.42$, $P_{27} = 6.87$, $P_{34} = 5.75$, $P_{35} = 4.86$, $P_{36} = 5.43$, $P_{37} = 5.14$, $P_{45} = 6.10$, $P_{46} = 6.88$, $P_{47} = 6.48$, $P_{56} = 5.76$, $P_{57} = 5.44$, $P_{67} = 6.11$, $P_{123} = 2.81$, $P_{124} = 3.58$, $P_{125} = 2.99$, $P_{126} = 3.37$, $P_{127} = 3.17$, $P_{134} = 2.66$, $P_{135} = 2.25$, $P_{136} = 2.52$, $P_{137} = 2.38$, $P_{145} = 2.82$, $P_{146} = 3.18$, $P_{147} = 2.99$, $P_{156} = 2.67$, $P_{157} = 2.52$, $P_{167} = 2.83$, $P_{234} = 3.80$, $P_{235} = 3.17$, $P_{236} = 3.58$, $P_{237} = 3.37$, $P_{245} = 4.04$, $P_{246} = 4.58$, $P_{247} = 4.30$, $P_{256} = 3.80$, $P_{257} = 3.58$, $P_{267} = 4.04$, $P_{345} = 2.99$, $P_{346} = 3.37$, $P_{347} = 3.18$, $P_{356} = 2.83$, $P_{357} = 2.67$, $P_{367} = 3.00$, $P_{456} = 3.58$, $P_{457} = 3.37$, $P_{467} = 3.81$, and $P_{567} = 3.18$.

The above uni-, bi-, and tri-component probabilities are used to compute the bounds on the series system probability by use of the theoretical formulas in Eqs. (8), (9), and (10). The same information is used to compute the bounds by LP. The LP formulation involves $2^7 = 128$ design variables, 7 equality constraints for the uni-component probabilities, 21 for bi-component probabilities, and 35 for tri-component probabilities.

Table 2. Bounds for Series System with Equal Component Probabilities

Bounds ($\times 10^{-3}$)		Lower	Upper
Unicomponent	Boole [Eq. (8)]	0.188	1.32
	LP	0.188	1.32
Bicomponent	KHD [Eq. (9)]	0.344–0.459	0.912–0.961
	LP	0.477	0.912
Tricomponent	Zhang [Eq. (10)]	0.605–0.631	0.809–0.833
	LP	0.631	0.796

Table 2 compares the theoretical bounds and the bounds by LP. For the bicomponent KHD and tricomponent Zhang bounds, all the possible $7! = 5,040$ ordering alternatives are considered and the range of results for each bound is shown with the best result in bold. As proven by Fréchet (1935), the unicomponent bounds by Boole are the best possible bounds when only the unicomponent probabilities are known. These are found to be identical to the unicomponent bounds by LP. For the bicomponent bounds, the best lower KHD bound is found to be smaller than the lower LP bound. This is an example of a case where the KHD formula does not provide the best possible bound for the given information, even when all the possible component ordering alternatives are considered. Note that there is a considerable dependence of the KHD bounds on the ordering of the components. For the tricomponent bounds, the best upper bound based on Zhang's formula is found to be greater than the upper LP bound. Thus the tricomponent Zhang formula does not provide the best possible bound for the given information, even when all the possible component ordering alternatives are considered. Note that there is considerable improvement in the bounds when tricomponent probabilities are included.

As a further example, consider the case where $r_i = \sqrt{\rho}$ in the DS correlation model, such that $\rho_{ij} = \rho$ for $i \neq j$ and $\rho_{ii} = 1$. This is the case of a series system with equiprobability and equicorrelated components. It is well known that this is the least favorable condition for bi- or tri-component bounds. Tables 3 and 4 list the computed bi- and tri-component probabilities and the corresponding LP, KHD, and Zhang bounds, respectively, for $\rho = 0.2, 0.4, 0.6, 0.8$, and 0.9 . Obviously the KHD and Zhang bounds are independent of the ordering of the components in this case. It is seen that in several cases (highlighted in bold) the KHD and Zhang bounds are inferior to the corresponding LP bounds.

Next, suppose that, for some reason, the joint probabilities involving component 1 are not available, i.e., P_{12} , P_{13} , P_{14} , P_{15} , P_{16} , P_{17} , P_{123} , P_{124} , P_{125} , P_{126} , P_{127} , P_{134} , P_{135} , P_{136} , P_{137} , P_{145} , P_{146} , P_{147} , P_{156} , P_{157} , and P_{167} are unknown. With the LP formulation, this problem is solved by simply removing six bicomponent probability constraints and 15 tricomponent probability constraints. For the correlation model of Table 2, the

Table 3. Bicomponent Bounds for Series System with Equal Component Probabilities and Correlations

ρ	$P_{ij} (1 \leq i < j \leq 7)$	Bounds ($\times 10^{-3}$)	
		LP	KHD [Eq. (9)]
0.2	4.11×10^{-7}	1.3062–1.3063	1.3062–1.3063
0.4	2.56×10^{-6}	1.26–1.30	1.26–1.30
0.6	1.10×10^{-5}	1.08–1.25	1.08–1.25
0.8	3.87×10^{-5}	0.606–1.08	0.552–1.08
0.9	7.20×10^{-5}	0.405–0.883	0.347–0.883

Table 4. Tricomponent Bounds for Series System with Equal Component Probabilities and Correlations

ρ	$P_{ijk} (1 \leq i < j < k \leq 7)$	Bounds ($\times 10^{-3}$)	
		LP	Zhang [Eq. (10)]
0.2	3.86×10^{-9}	1.3062– 1.3063	1.3062– 1.3103
0.4	1.68×10^{-7}	1.26– 1.27	1.26– 1.29
0.6	2.26×10^{-6}	1.12– 1.16	1.12– 1.20
0.8	1.72×10^{-5}	0.761– 0.927	0.761– 0.975
0.9	4.47×10^{-5}	0.514–0.720	0.493–0.746

bicomponent LP bounds are found to be $(4.43 \times 10^{-4}, 9.70 \times 10^{-4})$. These are wider than the LP bounds shown in the second row of Table 2 because of the missing information. The tricomponent LP bounds are found to be $(5.85 \times 10^{-4}, 8.77 \times 10^{-4})$, which are also wider than the corresponding LP bounds in the last row of Table 2 because of the missing information.

Finally, consider the case where inequality-type information is available. Suppose that for the equicorrelated system with $\rho = 0.9$, the inequalities $P_{ijk} \leq 5 \times 10^{-5}$ instead of equalities $P_{ijk} = 4.47 \times 10^{-5}$, $1 \leq i < j < k \leq 7$, are given. The LP solution for this case is obtained by simply replacing the equality constraints with the corresponding inequality constraints. The result is $(4.05 \times 10^{-4}, 7.57 \times 10^{-4})$. As expected, these bounds are tighter than the bicomponent LP bounds in Table 3, but wider than the tricomponent LP bounds in Table 4.

Example 2: Daniels' Parallel System

In 1945, H. E. Daniels investigated the reliability of a bundle of ideally brittle wires, as shown in Fig. 3 (Daniels 1945). He derived the exact failure probability of the system under the following conditions: (1) the wires are perfectly brittle and have identical, deterministic elastic moduli; (2) the wire strengths are statistically independent and identically distributed random variables, and (3) the load, L , is deterministic and is equally distributed among the surviving wires. Under these conditions, the “Daniels” system is a parallel system. In this example, we compute the uni- and bi-component bounds by LP and by the Boole and KHD formulas and compare them with the exact solution derived by Daniels.

Let $X_1, X_2, \dots, X_n$ denote the random strengths of the n wires in the bundle and $F(x)$ denote their common CDF. Also let $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ denote the set of ordered wire strengths. Note that, whereas X_i are statistically independent, $X_{(i)}$, in general, are statistically dependent. It is clear that the first wire will break if the event $E_1 = \{X_{(1)} \leq L/n\}$ occurs, the second wire will break if $E_2 = \{X_{(2)} \leq L/(n-1)\}$ occurs, and the i th wire, $1 \leq i \leq n$, will break if $E_i = \{X_{(i)} \leq L/(n-i+1)\}$ occurs. It follows that the Daniels system is a parallel system as defined in Eq. (4) with the component events E_i as just described.

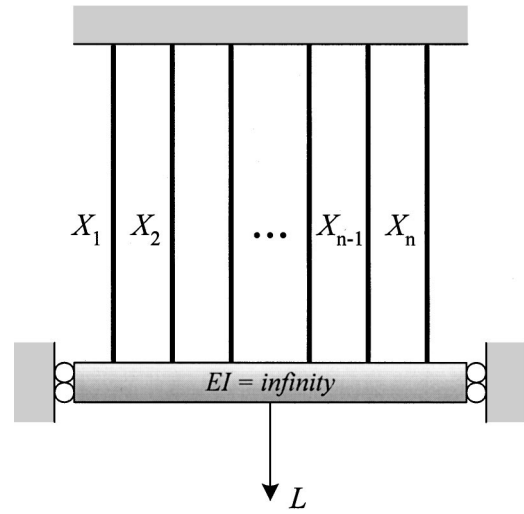


Fig. 3. Daniels system

Daniels (1945) used order statistics and a Taylor series expansion to derive the exact solution of the failure probability of the system as

$$P\left(\bigcap_{i=1}^n E_i\right) = (n!) \cdot \det \begin{bmatrix} b_n & \frac{b_n^2}{2!} & \frac{b_n^3}{3!} & \dots & \frac{b_n^{n-1}}{(n-1)!} & \frac{b_n^n}{n!} \\ 1 & b_{n-1} & \frac{b_{n-1}^2}{2!} & \dots & \frac{b_{n-1}^{n-2}}{(n-2)!} & \frac{b_{n-1}^{n-1}}{(n-1)!} \\ 0 & 1 & b_{n-2} & \dots & \frac{b_{n-2}^{n-3}}{(n-3)!} & \frac{b_{n-2}^{n-2}}{(n-2)!} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 & b_1 \end{bmatrix} \quad (19)$$

where $\det[\cdot]$ denotes the determinant of the matrix and $b_i = F(L/i)$.

To compute the unicomponent probabilities, we need the marginal distributions of $X_{(i)}$, $F_i(x) = P(X_{(i)} \leq x)$, and to compute the bicomponent probabilities, we need the bivariate distribution of $X_{(i)}$ and $X_{(j)}$, $F_{ij}(x, y) = P[(X_{(i)} \leq x) \cap (X_{(j)} \leq y)]$, for $1 \leq i < j \leq n$. Using order statistics (David 1970), one can show that

$$F_i(x) = \sum_{r=i}^n \binom{n}{r} F^r(x) [1 - F(x)]^{n-r} \quad (20)$$

$$F_{ij}(x, y) = \begin{cases} \sum_{r=i}^n \sum_{s=\max(0, j-r)}^{n-r} \frac{n!}{r!s!(n-r-s)!} F^r(x) [F(y) - F(x)]^s [1 - F(y)]^{n-r-s} & x < y \\ F_j(y) & x \geq y \end{cases} \quad (21)$$

Using these distributions, we have $P_i = P(E_i) = F_i[L/(n-i+1)]$ and $P_{ij} = P(E_i E_j) = F_{ij}[L/(n-i+1), L/(n-j+1)]$ for $1 \leq i < j \leq n$.

Using De Morgan's rule $\overline{\bigcap_i E_i} = \bigcup_i \overline{E_i}$, we convert the complement of the parallel system into a series system of the complementary component events. The KHD formula is then

$$\overline{F}_{ij}(x, y) = \begin{cases} \sum_{r=n-i+1}^n \sum_{s=\max(0, n-i+1-r)}^{n-r} \frac{n!}{r!s!(n-r-s)!} [1-F(y)]^r [F(y)-F(x)]^s F^{n-r-s}(x) & x < y \\ \overline{F}_i(x) & x \geq y \end{cases} \quad (22)$$

The above uni- and bi-component complementary probabilities are used to compute the KHD bounds on the probability of the complementary parallel system. These bounds are then subtracted from 1 to obtain the bounds on the parallel system failure probability.

As a numerical example, a Daniels system with 6 wires is considered. The wire strengths are assumed to have the Weibull distribution with CDF $F(x) = 1 - \exp(-\lambda x^\beta)$, $0 \leq x$, with $\lambda = 0.01$ and $\beta = 10$. Fig. 4 shows a comparison of Daniels' exact result (19), Boole's unicomponent bounds (11), the bicomponent KHD bounds (9), and the bicomponent bounds computed by LP. It is seen that the unicomponent bounds are too wide to be of practical use, especially when the sum of component probabilities is less than $(n-1)$, in which case the lower bound is simply zero. On the other hand, the bicomponent bounds by KHD and LP, which are practically coinciding for the range of load values considered, are fairly accurate for large load values (large probabilities). This example serves to demonstrate the applicability of the KHD and LP bounds to parallel systems. Of course, the LP bounds can be improved by considering tri- or higher-order component probabilities.

Example 3: Rigid-Plastic Structure as a General System

Consider a structural system consisting of an ideally elastic-plastic cantilever beam of moment capacity M , propped by an

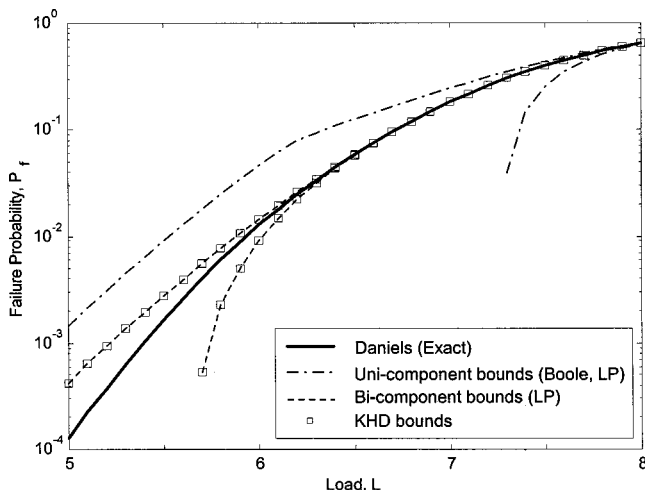


Fig. 4. Failure probabilities of Daniels system with six wires

applied to this series system. The corresponding unicomponent probabilities are $\overline{P}_i = P(\overline{E}_i) = 1 - P_i$ and the bicomponent probabilities are $\overline{P}_{ij} = P(\overline{E}_i \overline{E}_j) = \overline{F}_{ij}[L/(n-i+1), L/(n-j+1)]$, where $\overline{F}_{ij}(x, y) = P[(X_{(i)} > x) \cap (X_{(j)} > y)]$. Using order statistics, for $1 \leq i < j \leq n$, the latter is obtained as

ideally rigid-brittle bar of strength T (Fig. 5). A random load, X , is acting at the midpoint of the beam. As the load is applied, the failure of the system may occur along one of three possible paths, as shown in Fig. 6, each designating a cut set:

- Cut set 1: The bar fractures (event E_1), followed by the formation of a hinge at the fixed end of the cantilever beam (event E_2).
- Cut set 2: A hinge forms in the fixed end of the beam (event E_3), followed by another hinge in the midspan (event E_4).
- Cut set 3: A hinge forms in the fixed end of the beam (event E_3), followed by fracture of the bar (event E_5).

Thus the system failure event is described as

$$E_{\text{system}} = E_1 E_2 \cup E_3 E_4 \cup E_3 E_5 \quad (23)$$

Performing elementary structural analysis for the intact structure as well as each of its damaged states, one can easily relate the component events to the load and capacity variables as follows: $E_1 = \{T - 5X/16 \leq 0\}$, $E_2 = \{M - LX \leq 0\}$, $E_3 = \{M - 3LX/8 \leq 0\}$, $E_4 = \{M - LX/3 \leq 0\}$, and $E_5 = \{M + 2LT - LX \leq 0\}$.

The above is a minimum cut-set formulation of the problem. In such a formulation, the cut sets are neither disjoint nor statistically independent. A formulation using disjoint cut sets would require the introduction of additional component events to ensure that each cut set represents a distinct failure path. For example, to assure that the bar fractures before a hinge forms in the fixed end of the beam, we must add the component event $E_6 = \{6LT - 5M \leq 0\}$ to the first cut set. To describe the second distinct path of failure, i.e., the formation of a hinge in the fixed end of the beam followed by a hinge in the midspan, we must add $\overline{E}_6$ and

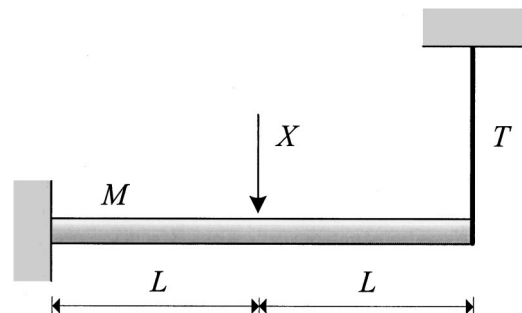


Fig. 5. Cantilever beam-bar system

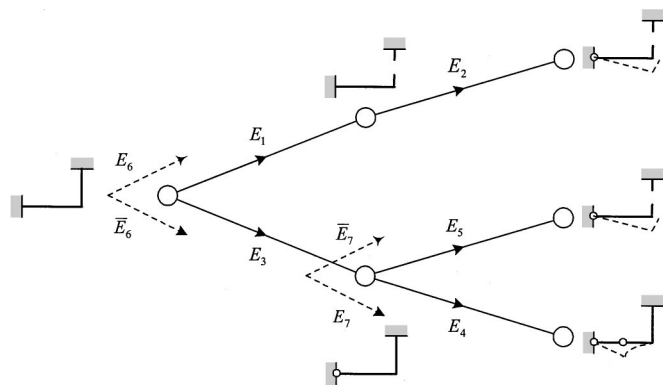


Fig. 6. Failure paths of cantilever-bar system

$E_7 = \{M - LT \leq 0\}$ to the second cut set. Finally, to describe the third distinct path of failure, i.e., formation of a hinge in the fixed end of the beam followed by fracture of the bar, we must add $\bar{E}_6$ and $\bar{E}_7$ to the third cut set. The disjoint cut set formulation, $E_{\text{system}} = E_1 E_2 E_6 \cup E_3 E_4 \bar{E}_6 E_7 \cup E_3 E_5 \bar{E}_6 \bar{E}_7$, has the advantage that the probabilities from the individual cut sets can be computed separately and then added to obtain the system probability. However, from the viewpoint of computing bounds by LP, the minimum cut-set formulation is advantageous, since it involves the smallest number of component events, a number which directly controls the size of the LP problem. In the following, we will only consider the minimum cut-set formulation (23).

For the sake of simplicity, we assume the random variables M , T , and X have the normal distribution. In that case, the limit-state functions defining component events are jointly normal. Thus the unicomponent probabilities can be computed by the standard normal CDF, the bicomponent probabilities can be computed by the standard bivariate normal CDF, and so forth. With this distribution assumption, no approximations are introduced in the computation of the component probabilities, thus allowing us to examine the exact bounds computed by LP. It is noted that the exact solution of the above problem requires computation of the multinormal probability distribution for dimensions 1–5.

As a numerical example, we consider the case where $L=5$ and M , T , and X are uncorrelated normals with means $\mu_M=150$, $\mu_T=1,000$, and $\mu_X=110$ and standard deviations $\sigma_M=30$, $\sigma_T=300$, and $\sigma_X=20$, respectively. The uni- and bi-component probabilities for this case are listed in Table 5. The LP bounds based on the unicomponent probabilities alone are found to be $(0, 9.24 \times 10^{-3})$. The bicomponent LP bounds are found to be $(7.74855152 \times 10^{-3}, 7.74855155 \times 10^{-3})$. The latter bounds show a difference only in the ninth digit. To verify these calculations, Monte Carlo simulations are conducted. After 10^7 simulations, the system failure probability is estimated as 7.770×10^{-3} with a coefficient of variation of 0.003,57. The one-standard deviation interval of the simulation estimate is $(7.742 \times 10^{-3}, 7.798 \times 10^{-3})$, which includes both the lower and upper bicomponent LP bounds. Recalling that the exact solution of the problem requires the joint probabilities of all five component events, it is remarkable that for this system the bicomponent LP bounds essentially provide the exact solution.

Summary and Conclusions

It is shown that LP can be used to obtain the best possible bounds on a system probability for any level of information on marginal

Table 5. Uni- and Bi-Component Probabilities for Rigid-Plastic Structure

P_{ij}	$j=1$	2	3	4	5
$i=1$	2.13×10^{-3}	9.39×10^{-4}	3.68×10^{-5}	2.60×10^{-5}	6.85×10^{-5}
2		2.28×10^{-1}	9.27×10^{-3}	6.83×10^{-3}	2.73×10^{-4}
3			9.27×10^{-3}	6.83×10^{-3}	2.41×10^{-4}
4		$P_{ij} = P_{ji}$		6.83×10^{-3}	2.27×10^{-4}
5					2.73×10^{-4}

or joint component probabilities. The information can be in the form of equality or inequality expressions of component probabilities. The method is applicable to any system state defined as a logical expression of component states. This includes series and parallel systems, as well as general systems that are characterized in terms of cut sets or link sets.

The LP methodology is demonstrated by its application to three structural reliability problems. For a truss structure modeled as a series system, it is shown that the commonly used KHD and Zhang bounds may not be the best bounds for the level of information that is utilized. This example also shows how the LP bounds can be obtained when the information on component probabilities is incomplete or in the form of inequalities. The second example involves a Daniels system consisting of an ideally brittle bundle of wires. This is a parallel system, for which an exact solution is available. The uni- and bi-component probability bounds using LP are computed and compared with the exact solution. It is shown that the KHD bounds, which are normally used for series systems, can also be used for parallel systems. For this example, the KHD bounds practically coincide with the corresponding LP bounds. The third example is for a structure with both ductile and brittle members. The structure is modeled as a system with three cut sets and five components. The uni- and bi-component bounds are computed by LP and compared with a simulation result, as no theoretical bounding formula for general systems is available. The bicomponent LP bounds for this example are found to be extremely narrow.

The only shortcoming of the LP approach for computing the system probability bounds is that the size of the LP problem grows exponentially with the number of component states. With rapidly increasing computer speed and memory capacity, and with publicly available highly advanced LP algorithms and codes, this shortcoming is not critical, at least for systems with no more than 17 components. Some ideas are suggested for reducing the size of the LP problem for systems with large number of components.

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Discussion of "Bounds on System Reliability by Linear Programming" by Junho Song and Armen Der Kiureghian

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The authors have produced a useful paper in the field of structural reliability bounds. They have applied linear programming techniques and obtained optimal bounds for a few examples in structural mechanics. However the authors have inadvertently omitted a few useful publications on probability bounds (Ramachandran 1984, 1990, 2001; Greig 1992), which contain methods of obtaining near-optimal bounds without linear programming.

Kounias and Marin (1976) attempted to use linear programming techniques to get optimal second-order bounds. They were trying to find the coordinates of the vertices explicitly and observed that it would be very difficult, if not impracticable, to find all the vertices when the number of events is greater than four, and they indicated that if the numerical values of all P_i , P_{ij} are available, a computer solution is possible for a larger number of events. The authors appear to have resurrected this approach with modifications to include third-order (third degree) components (tri-sections). The authors have also noticed, as Kounias and Marin did, that there is a limit on the number of events that can be handled by linear programming techniques. This problem (except the second-order upper bound) is a NP-Hard problem, and closed form solutions are not easy to find. Hence, computer requirements will become excessive with an increased number of events.

The discussor produced an elegant and computationally efficient method for the evaluation of second-order upper bound (Ramachandran 2001). For completeness, the method is outlined here. Select any event k and choose the largest P_{kj} for all $j=1, n$ and $j \neq k$. Let it be P_{kj} . Then, choose the largest P_{ki} , P_{ji} for all $i=1, n$ and $i \neq j$, $i \neq k$ and let it be P_{ki} . Continue the process of finding the largest P_{ki} , P_{ji} , P_{li} for all $i=1, n$ and $i \neq j$, $i \neq k$, $i \neq l$. Repeat the process until $(n-1)$ bisections are found. For the Example 1 of the authors' paper, this algorithm would have sequentially selected P_{12} (if the starting event is 1, we can start with any event), P_{24} , P_{28} , P_{27} , P_{25} , P_{13} , and the optimum second-order upper bound P_{2U} would be $[7 \times 18.8 - (5.13 - 7.79 - 7.42 - 6.87 - 6.47 - 6.08)] \times 10^{-5} = 91.24 \times 10^{-5}$.

The discussor also produced in the same papers an algorithm to select a near-optimal second-order lower bound. The process is again very simple and computationally very efficient and produced optimal bounds in all cases tested. Choose the largest P_{i1} . Then choose the largest $[P_{i2} - P_{i1i2}]$ for all $i2=1, n$, $i2 \neq i1$. The next step is to choose the largest $[P_{i3} - P_{i1i3} - P_{i2i3}]$, $i3=1, n$, $i3 \neq i1$, $i3 \neq i2$. Repeat the process until the quantity $[P_{in'} - P_{i1in'} - P_{i2in'} - P_{i3in'} \dots - P_{i(n'-1)in'}]$ where $n' \leftarrow n$ is less than zero. The lower bound is the sum of all these positive contributions. Again,

for Example 1 in the authors' paper, the discussor's algorithm would select lower bound P_{2L} as

$$P_{2L} = P_1 + P_3 - P_{13} + P_5 - P_{15} - P_{35} + P_7 - P_{17} - P_{37} - P_{57} \\ = 45.97 \times 10^{-5}$$

The authors' second-order lower bound 47.7×10^{-5} is definitely better than that produced by the discussor's algorithm. The discussor knows the existence of this problem and accepts that it is possible to improve the discussor's algorithm to get a higher (better) lower bound, and the discussor's findings will be published soon. But, the extra amount of computation required to get a cosmetic improvement on the second-order lower bound may not be worthwhile in structural reliability calculations (see later for more discussion). However, it would be helpful if the authors could let us know the values of all P_i and P_{ij} that yielded their optimal lower bound.

The discussor's algorithm for third-order bounds produced a value of 63.13×10^{-5} for lower bound P_{3L} . This lower bound is same as the optimal bound produced by the authors, 63.1×10^{-5} . It is interesting to note that the original numbering of events without any reordering would also have produced this optimal bound. The discussor's algorithm would give an upper bound of $P_{3U} = 80.11 \times 10^{-5}$. This upper bound is slightly higher than the authors' upper bound, 79.6×10^{-5} . The discussor's third-order upper bound uses two bisections and a single trisection only. It has been shown (Ramachandran 1984; 2001; Greig 1992) that more bisections and corresponding trisections may be included to get better bounds.

From this discussion it should be clear that the discussor's simple algorithms appear to give optimal or near-optimal bounds at a fraction of the cost of using linear programming techniques. However, the discussor's algorithms at present cannot deal with probabilities that do not have fixed values, unlike the authors' linear programming method. Readers of this discussion should note this point.

In summarizing, the discussor would like to raise the following issues. In engineering mechanics problems, the values of P_i and P_{ij} are usually obtained by making a number of approximations (linearization, approximate correlation, matrix etc). Furthermore, bounds on structural reliability give only an indication of the range for failure probabilities and do not give absolute values. Under these circumstances, is it prudent to look for optimal bounds in structural reliability? Of course it is academically exciting and challenging to get optimal bounds for a given set of P_i , P_{ij} , and P_{ijk} . Linear programming techniques, as authors have noticed, have limitations on the number of events that can be handled, even with reasonably large computers. These observations suggest that simple algorithms that can give near-optimal bounds are worthy of being produced and should be incorporated into any reliability package.

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Closure to "Bounds on System Reliability by Linear Programming" by Junho Song and Armen Der Kiureghian

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The writers appreciate the discussor's interest in the paper. The discussor questions the use of linear programming (LP) in estimating bounds on structural system reliability on the grounds that it may not be prudent to seek the narrowest possible bounds by solving an LP problem when approximations are often involved in computing component probabilities. The discussor advocates simple existing formulas that can provide near-narrowest possible bounds with less effort.

As pointed out in the paper, theoretical bounding formulas can provide bounds for series and parallel systems when complete sets of equality constraints on marginal or joint component probabilities are available. The paper does not deny the usefulness of those formulas, including that of the discussor (Ramachandran 1984, 1990) which we did not cite, for the limited class of systems under the specific type of constraints. Rather, it proposes a general methodology that not only guarantees the narrowest possible bounds without ordering dependency but is also generally applicable to any system (not only series and parallel) and incorporates a much broader class of constraints on component probabilities. We note that the discussor is only concerned with series systems.

The discussor argues that the approximations involved in computing component probabilities make it unnecessary to pursue the higher accuracy that LP bounds could achieve. On the contrary, the writers find the LP bounds to be even more useful in that case. The approximation involved in computing component probabilities often leads to information in the form of incomplete sets of component probabilities or inequality constraints, i.e., bounds on component probabilities. In contrast to the theoretical bounding formulas, the LP formulation can easily incorporate this type of information, as shown by Example 1 in the paper. Furthermore, the writers believe that the existence of errors in input data should not discourage the development of an accurate analysis method, especially when the method helps to avoid further assumptions or approximations in incorporating the input.

In many cases, the failures of structural systems are not defined as either series or parallel systems. Unlike the theoretical formulas derived for the specific classes of series and parallel

systems, the LP approach is uniformly applicable to any type of system. The difference in analysis is only in the composition of the coefficient vector $\mathbf{c}$ in Eq. (12). Once set up as an analysis package, one can use the LP approach uniformly for all types of systems.

Recently, the writers have shown that the proposed LP approach can be used to systematically identify the critical components and cut sets of a system (Song and Der Kiureghian 2004a). Once the bounds on the system failure probability are obtained by LP, simple algebraic calculations yield well-known importance measures for components and cut sets. Thus, the LP formulation provides possibilities beyond computing bounds on system probability.

As we have acknowledged in the paper, the size of the LP problem grows exponentially with the number of components. The writers believe that with rapidly advancing speed and capacity of computers this purely computational issue may not be a hindrance in application of the LP approach to large-scale system reliability problems. Furthermore, there already exist advanced LP algorithms, such as the column generation method (Jaumard et al. 1991), which can efficiently handle large-size LP problems. Finally, the authors have developed a method to reduce the size of the LP problem by treating sets of components as "super-components." In effect, in this method, the large LP problem for the system is decomposed into a number of smaller LP problems at the expense of a slight relaxing of the bounds (Song and Der Kiureghian 2004b).

The discussor requested the values of all the unicomponent and bicomponent probabilities that yielded the best lower bound in Example 1. Those values are already presented in the paper. Since the LP bounds method provides the best possible bounds independently of ordering choices, there is no need to find or present the ordering choice yielding the best bounds.

In summary, the paper proposes a general methodology that has numerous merits, potentials, and byproducts, which are not offered by the existing bounding formulas. These include (a) the ability to compute bounds for general (nonseries or parallel) systems; (b) the ability to handle all types of information on component probabilities, including incomplete or bounding information; (c) the ability to obtain the narrowest possible bounds for any given information; and (d) the facility to compute importance measures for components and cut sets. While the computational issues of large-size LP problems are being overcome by rapidly growing computing technology and special LP algorithms, the use and further development of the proposed method should not be discouraged. The authors hope this discussion clarifies the main points of the paper and helps the structural reliability community appreciate the numerous advantages of the LP bounds methodology. We thank the discussor for the opportunity provided to clarify these issues.

Errata

The following corrections should be noted:

1. In Eq. (22) on page 634, the index r of the first summation should vary from $(n-j+1)$ to n instead of $(n-i+1)$ to n .
2. In Example 3, on page 635, the values for the means and standard deviations of M , T , and X should be $\mu_M=1,000$, $\mu_T=110$, $\mu_X=150$, $\sigma_M=300$, $\sigma_T=20$, and $\sigma_X=30$.

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